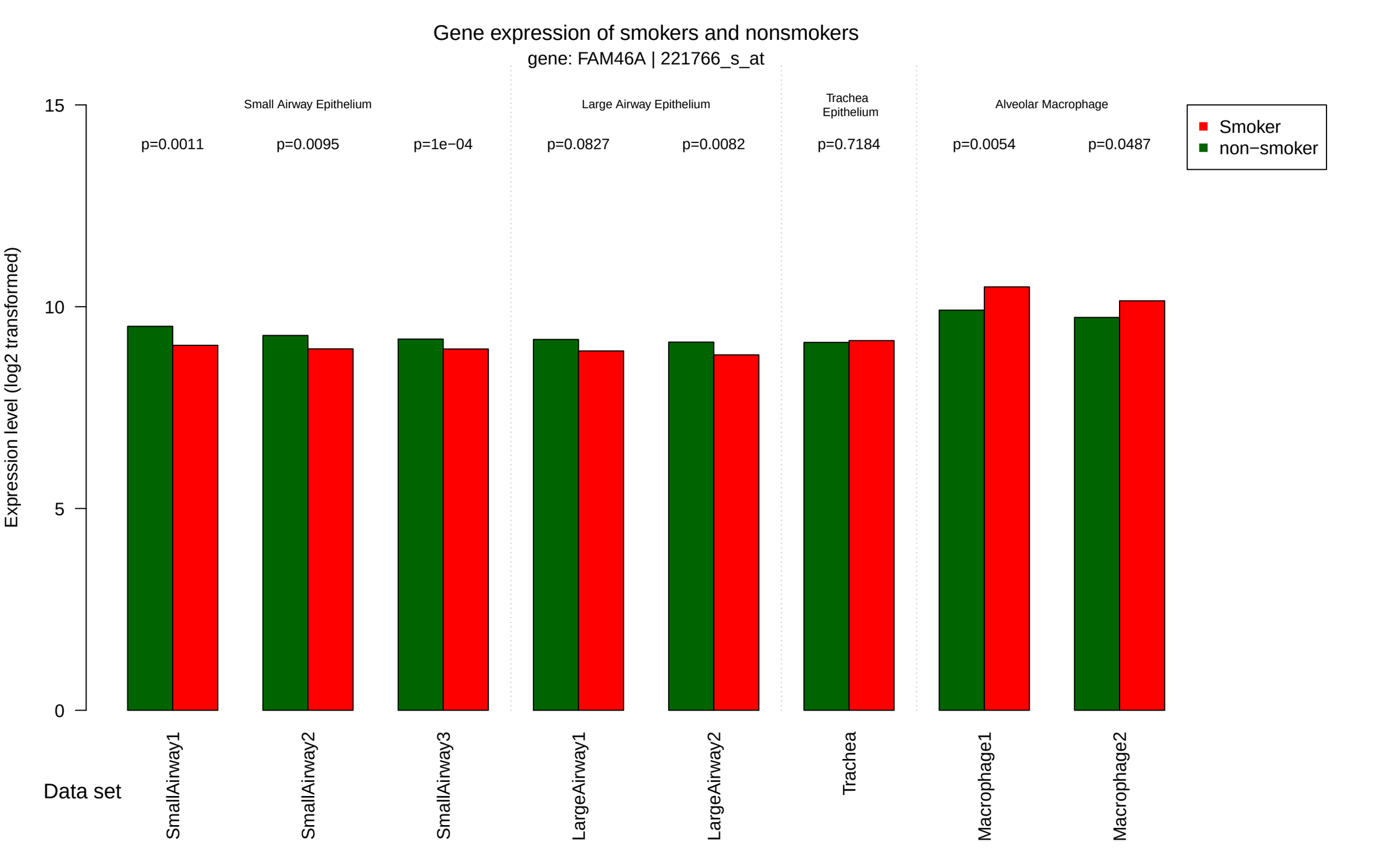
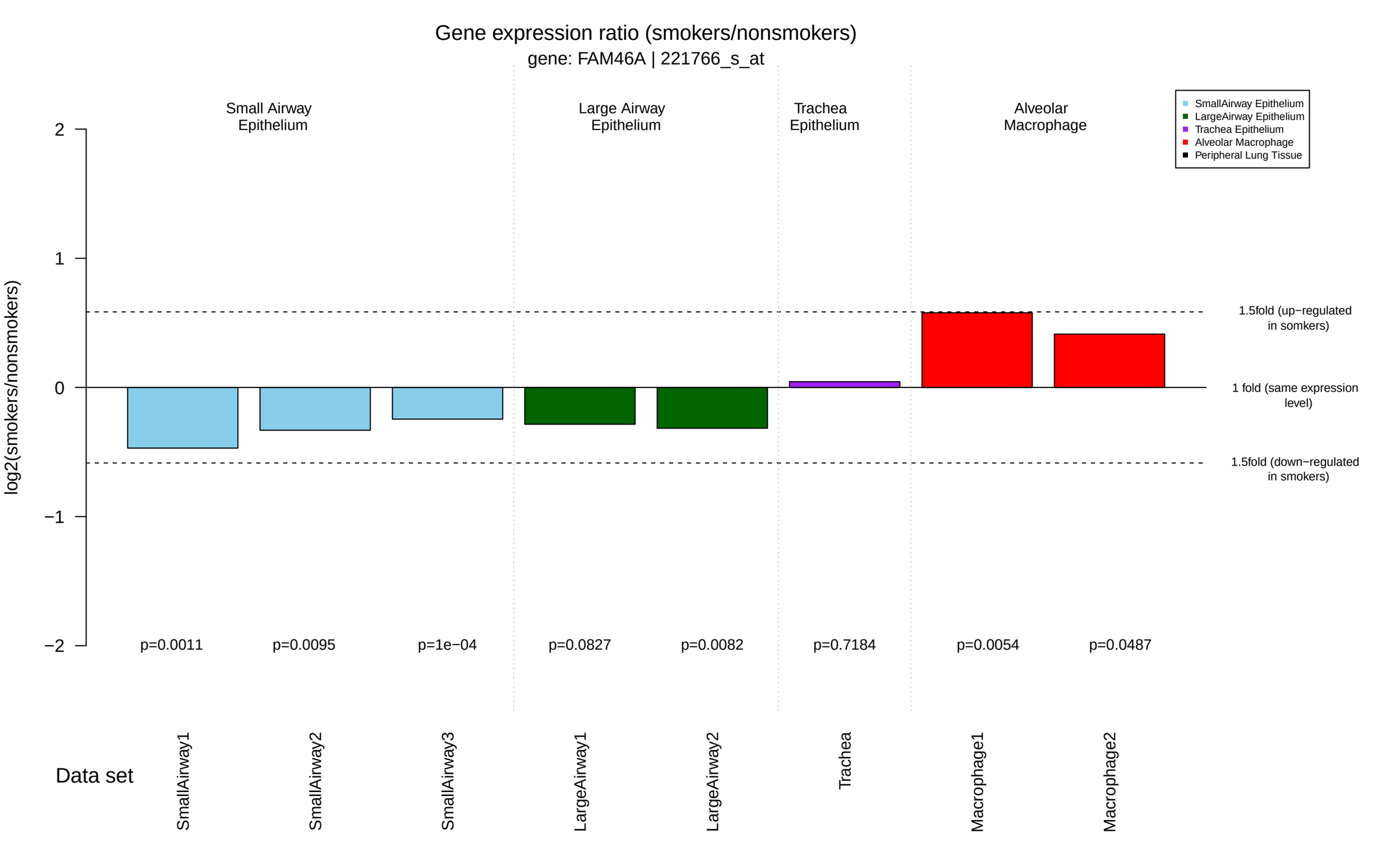
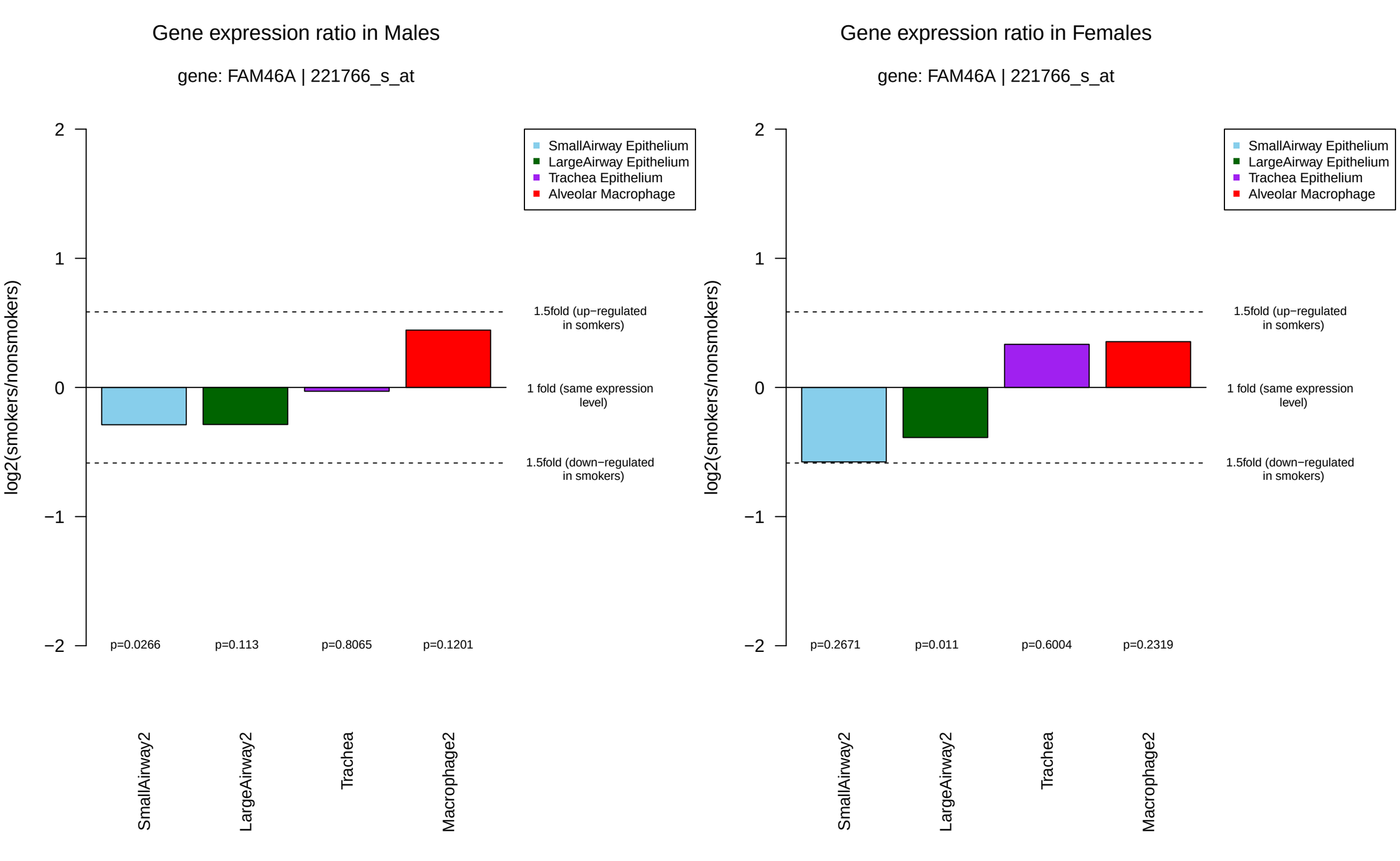
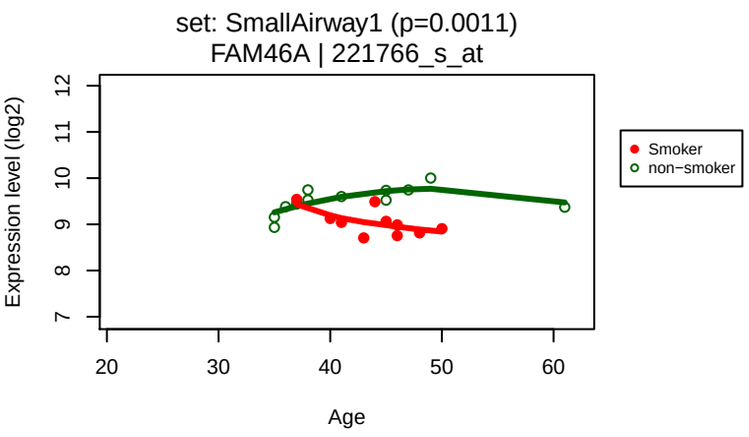
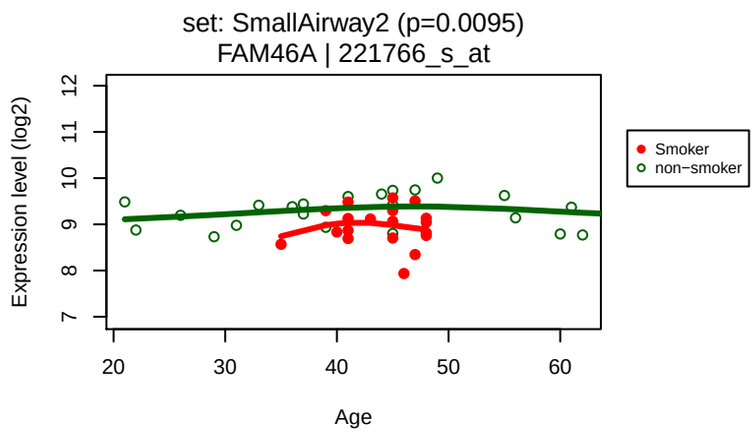
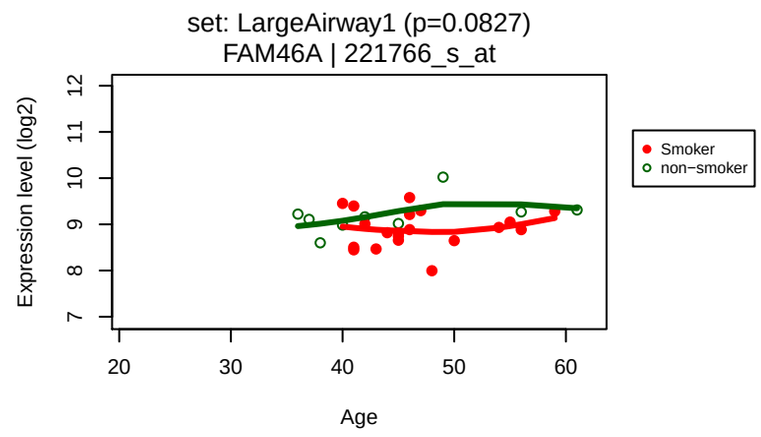

Gene expression of smokers and nonsmokers
gene: FAM46A | 221766_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0011
p=0.0095
p=1e−04
p=0.0827
p=0.0082
p=0.7184
p=0.0054
p=0.0487
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

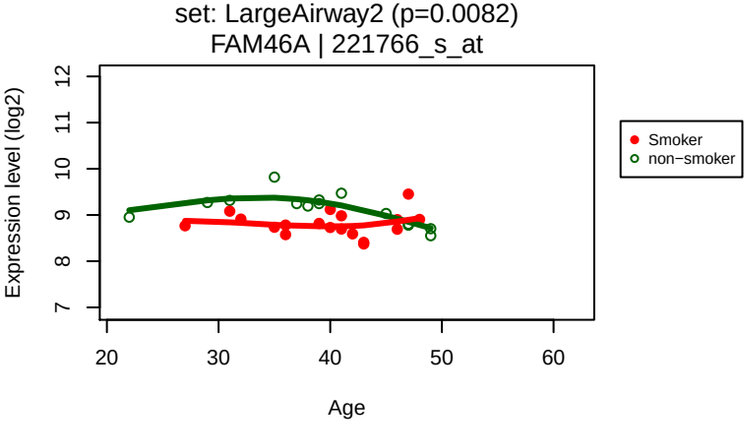

Gene expression ratio (smokers/nonsmokers)
gene: FAM46A | 221766_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0011
p=0.0095
p=1e−04
p=0.0827
p=0.0082
p=0.7184
p=0.0054
p=0.0487
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

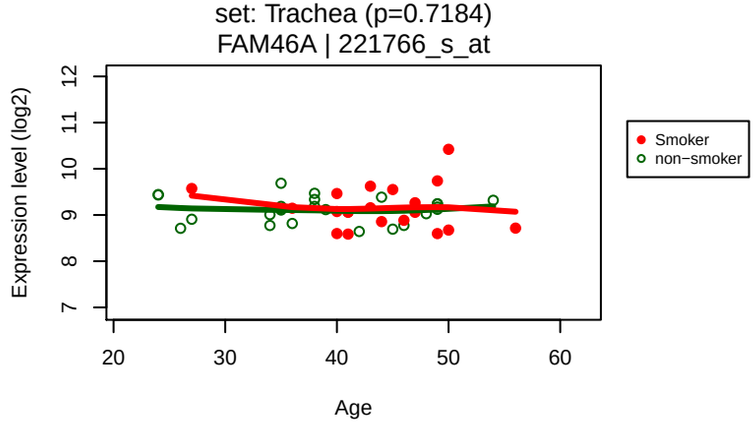

Gene expression ratio in Males
gene: FAM46A | 221766_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0266
p=0.113
p=0.8065
p=0.1201
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: FAM46A | 221766_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2671
p=0.011
p=0.6004
p=0.2319
SmallAirway2
LargeAirway2
Trachea
Macrophage2

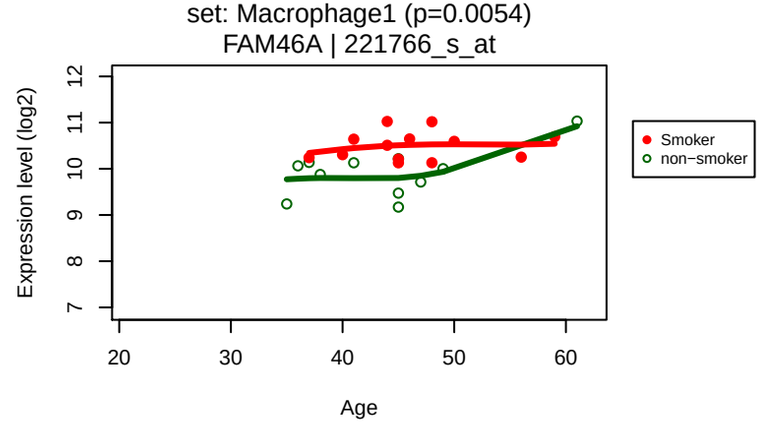

set: SmallAirway1 (p=0.0011)
FAM46A | 221766_s_at
Expression level (log2)
Age
Smoker
non−smoker

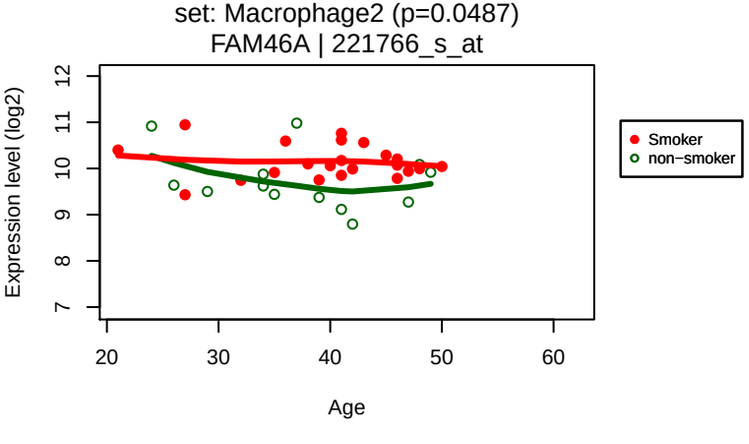

set: SmallAirway2 (p=0.0095)
FAM46A | 221766_s_at
Expression level (log2)
Age
Smoker
non−smoker

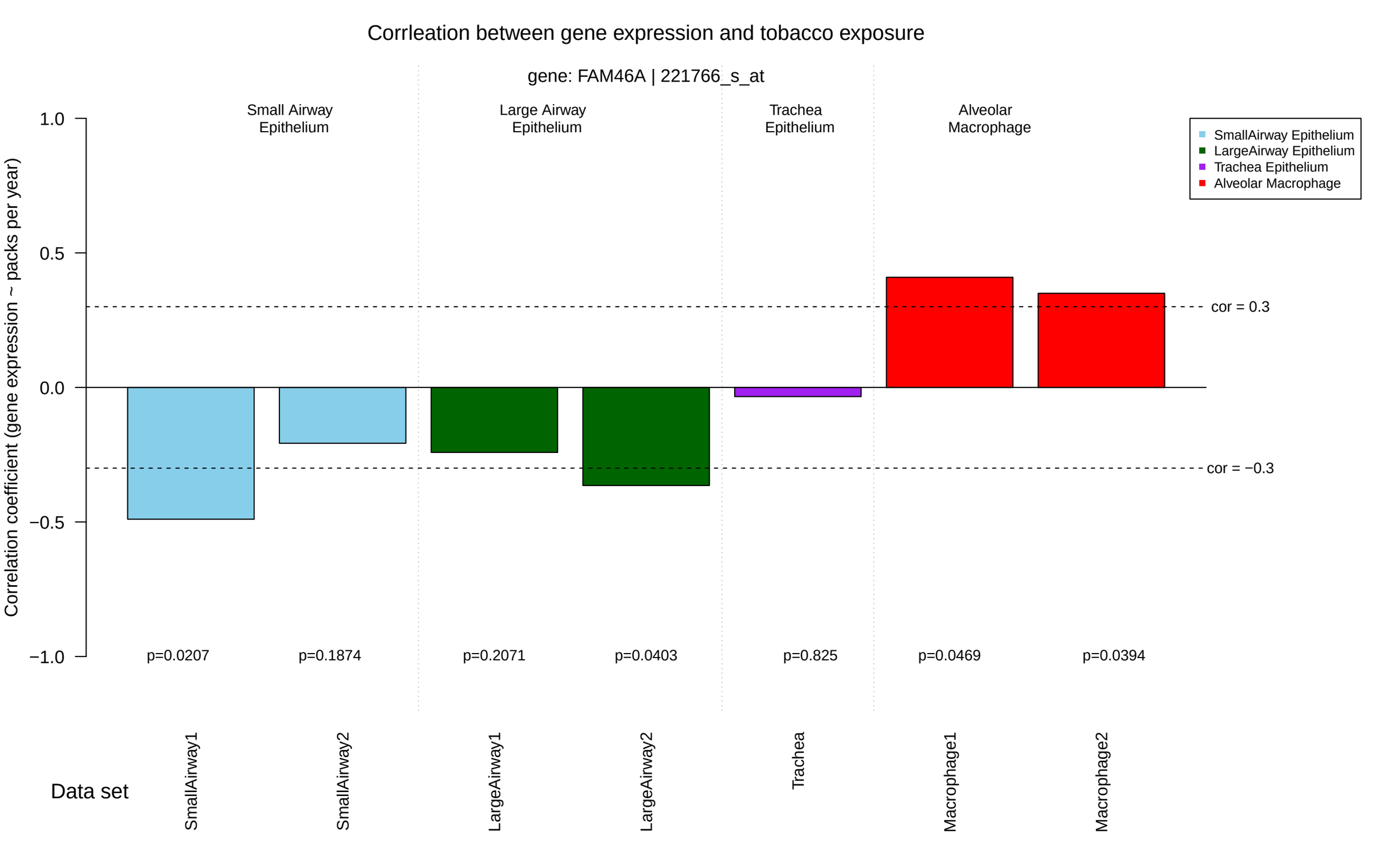

set: LargeAirway1 (p=0.0827)
FAM46A | 221766_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0082)
FAM46A | 221766_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7184)
FAM46A | 221766_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0054)
FAM46A | 221766_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0487)
FAM46A | 221766_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: FAM46A | 221766_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0207
p=0.1874
p=0.2071
p=0.0403
p=0.825
p=0.0469
p=0.0394
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2